

Fakturama Image-to-Cash — Design, in brief

Ahmed Islam Farouk Abbas · TJM Labs take-home, Part 1 · full version in `DESIGN.md`

One order image in; a saved, verified Order and its linked Invoice out. No hardcoded coordinates, nothing assumed about window size, theme or DPI.

The interesting part is not clicking. It is deciding, at each step, whether what I am looking at is the thing I meant — and stopping when I cannot tell.

Shape. `extract` → `flow` → `driver`. `extract` is pure, so it ships and is tested before a Windows VM exists. `flow` is the spec transcribed, step numbers in the comments. `driver` is the only module that knows what a window is.

Finding controls

Fakturama is Eclipse/SWT: `UIA` sees a real control tree, but almost nothing carries an `AutomationId`, names are locale-derived, and the two icons beside `Addresses` — where the spec insists on the *upper* one — have no name at all. So, first unambiguous hit wins:

TIER	METHOD
T1	scoped search — resolve the window first, then search inside it
T2	anchor-relative walk — a named landmark, then the <code>nth</code> child
T3	tooltip / description / enabled state
T4	screenshot the anchor's own <code>UIA</code> rectangle , ask a model which, click by handle

T2 is the answer to "no hardcoded coordinates": ordering is a property of the widget hierarchy, not of the screen, read from the live tree at call time. All 104 selectors live in one dict, so a locale or version change is a data edit.

Reading the image

LLM vision constrained to a JSON schema, because it returns *table structure* — which column is `Disc.`, which value is `BILLING` — rather than a bag of words whose geometry I then rebuild. Local OCR is the last tier and needs no account at all.

Then the document is checked against itself, so correctness is decidable without a human: $qty \times unit_net \times (1 - disc) = line_net$ per line, lines summing to `570.00`, VAT `108.30`, gross `678.30`. A break means a misread digit: re-prompt once, then halt. Money is `Decimal`, never `float`. The limit is honest — a misread SKU passes all four, and is caught later by Fakturama's own selector finding no match.

Verifying, and stopping

- **Selection is the existence check.** Never query the database — ask the Order's own picker. A record not selectable *from the Order* is correctly absent.
- **Saving is confirmed by re-selection.** The proof a Debtor persisted is that the Order's picker now finds it.

Ambiguity — more than one exact candidate, or one conflicting with the source — raises one `AmbiguityHalt` carrying the control, the query, the candidates and a screenshot, and exits non-zero. Guessing between two Debtors is the one failure this must never produce. Idempotency falls out: every stage is `check` → `create` → `re-select`.

Tradeoffs

`UIA` first, vision only as a tiebreak: fast and deterministic, but Windows-only. Anchor-relative resolution survives resize, DPI and theme, and breaks only if the hierarchy is restructured. The arithmetic gate is free and decidable, and blind to errors that preserve the identities. Halting on ambiguity means no silently wrong records, at the cost of needing a human for the interesting cases.